Notebook: myutils.py – Utility Functions

This module contains a collection of reusable utility functions developed for use across all Jupyter notebooks in the project. The primary objective was to standardize the data analysis workflow, reduce code duplication, and ensure a consistent style for data visualization and reporting.

Main implementation steps:

- Configured the required libraries for data analysis and visualization (**Pandas, NumPy, Matplotlib**) and established a unified plotting style, including a consistent color palette, labels, and grid settings.
- Developed functions for automated analysis of **categorical** and **numerical** variables, including descriptive statistics, identification of the most frequent values, and missing value analysis.
- Implemented a comprehensive dataset reporting function that provides information about dataset dimensions, memory usage, missing values, and key descriptive statistics.
- Developed reusable visualization functions for:
  - Distribution of categorical variables (counts and percentages);
  - Analysis of numerical variables using histograms and boxplots to identify distributions and detect outliers.
- Implemented a function to calculate **Conversion Rate** across selected grouping variables, enabling the evaluation of performance across different categories and traffic sources.

Result

A unified data analysis module was developed and applied throughout the entire project. It automated repetitive analytical tasks, ensured a consistent presentation of results, and significantly improved the efficiency of the overall data analysis process.

Notebook: 01_contacts_prepare – Contact Data Preparation and Cleaning

This notebook covers the complete process of preparing and cleaning the original contact dataset for further analysis and dashboard development.

Main implementation steps:

- Imported the required libraries and integrated the **myutils.py** module for automated data analysis and visualization.
- Loaded the raw Excel contact dataset and performed an initial assessment of the dataset structure, including its dimensions, columns, and data types.
- Conducted an initial data quality assessment using the **display_report()** function, including the analysis of missing values as well as categorical and numerical features.
- Converted the **Created Time** and **Modified Time** fields to the **datetime** format and created additional date-only columns for future time-series analysis.
- Processed missing values:
    - Replaced missing values in the **Manager** field with **"Unknown"**.
    - Verified that the primary identifier (**Id**) contained no missing values.
- Removed duplicate records based on the **Id** field to ensure contact uniqueness.
- Optimized data types:
    - Converted the **Id** field to the **string** data type.
    - Converted the **Manager** field to the **category** data type to reduce memory usage and improve grouping performance.
- Created a new analytical feature, **modification_delay_days**, representing the number of days between contact creation and the most recent modification.
- Performed a second data quality assessment on the cleaned dataset to validate the preprocessing results.
- Created visualizations for:
    - Contact distribution by manager;
    - Monthly trend of newly created contacts, highlighting periods of peak activity.
- Saved the final cleaned dataset in **.pkl** format for subsequent notebooks within the project.

Result

A cleaned and optimized contact dataset was produced, free of duplicates, with correctly assigned data types and additional analytical features. The dataset is fully prepared for further analysis and integration with the remaining project datasets.

Notebook: 02_calls_prepare – Call Data Preparation and Cleaning

This notebook focuses on preparing and cleaning the call dataset for further analysis. The objective was to improve data quality, standardize data formats, generate additional analytical features, and prepare the dataset for integration with other project data sources.

Main implementation steps:

- Imported the required libraries and connected the **myutils.py** utility module to ensure a standardized analysis workflow.
- Loaded the raw call dataset from the original Excel file and performed an initial assessment of the dataset structure, including its dimensions, columns, data types, and missing values.
- Conducted a comprehensive data quality assessment using the **display_report()** function.
- Converted the **Call Start Time** field to the **datetime** format to enable time-based analysis.
- Generated additional analytical features from the call timestamp:
  - **call_date** – date of the call;
  - **call_hour** – hour of the call;
  - **day_of_week** – weekday on which the call occurred.
- Verified the dataset for missing values and duplicate records to ensure data integrity.
- Created visualizations to analyze:
  - Distribution of calls by weekday;
  - Distribution of calls by hour of the day.
- Saved the cleaned dataset in **.pkl** format for use in subsequent stages of the project.

Result

A standardized and cleaned call dataset was created, featuring optimized data types and additional time-based attributes. The dataset is fully prepared for call activity analysis, integration with CRM data, and dashboard development.

Notebook: 03_spend_prepare – Marketing Spend Data Preparation and Cleaning

This notebook covers the preprocessing and validation of the marketing spend dataset. The objective was to standardize the data structure, prepare time-based features, verify data quality, and generate a clean dataset for marketing performance analysis.

Main implementation steps:

- Imported the required libraries and connected the **myutils.py** utility module.
- Loaded the raw marketing spend dataset from the original Excel file.
- Performed an initial assessment of the dataset, including its structure, dimensions, data types, and missing values.

- Converted the **Date** field to the **datetime** format.
- Created a new **year_month** feature to support monthly trend analysis and KPI reporting.
- Validated the key numerical metrics:
    - **Impressions**
    - **Clicks**
    - **Spend**
- Checked the dataset for missing values, incorrect data types, and duplicate records.
- Generated descriptive statistics and visualizations to evaluate the distribution of marketing metrics.
- Saved the cleaned dataset in **.pkl** format for further analysis and integration with CRM data.

Result

A validated and standardized marketing spend dataset was created, containing consistent date formats and optimized data types. The dataset is fully prepared for campaign performance analysis, KPI calculations, and integration with sales and CRM data.

Notebook: 04_deals_prepare – Deal Data Preparation and Feature Engineering

This notebook covers the complete preprocessing, cleaning, standardization, and feature engineering of the CRM deals dataset. Since the deals table serves as the project's primary analytical dataset, particular attention was given to data quality, business logic implementation, and the creation of calculated metrics required for further analysis and dashboard development.

Main implementation steps:

- Imported the required libraries and connected the **myutils.py** utility module.
- Loaded the raw CRM deals dataset from the original Excel file.
- Performed an initial assessment of the dataset structure, including dimensions, data types, missing values, duplicate records, and descriptive statistics.
- Converted the primary identifier (**Id**) and **Contact Name** to the **string** data type to ensure data consistency during dataset integration.
- Removed empty rows and duplicate records to improve overall data quality.
- Standardized categorical variables by:
    - Normalizing **Lost Reason** values;
    - Replacing missing values in **Campaign**, **Content**, and **Term** with **"not set"**;
    - Standardizing **Payment Type** values;
    - Cleaning and normalizing **Product** names;
    - Standardizing **Stage** values using unified business rules.

- Converted all date fields to the **datetime** format for time-based analysis.
- Processed numerical fields by converting monetary values and payment-related columns into consistent numeric formats.
- Implemented custom logic to calculate **SLA Minutes**, ensuring reliable measurement of processing time across deals.
- Created the **Is Paid** indicator based on the current deal stage.
- Standardized the **Quality** field and created the corresponding **Quality Score** numerical variable for analytical purposes.
- Corrected inconsistencies in **Closing Date** records and validated chronological relationships between important dates.
- Implemented business logic to classify deals into payment categories:
    - **One Payment**
    - **Recurring Payments**
    - **No Payment**
- Developed custom business logic to calculate **Revenue** based on payment type, course duration, initial payment amount, and completed months of study.
- Calculated the total number of **Transactions** for each deal according to the implemented payment model.
- Performed a final validation of the cleaned dataset and generated descriptive statistics and visualizations for key analytical variables.
- Saved the fully processed dataset in **.pkl** format for subsequent analysis and dashboard development.

Result

A fully cleaned, standardized, and business-ready CRM deals dataset was created. The notebook implemented all required business rules, generated key analytical metrics such as **Revenue**, **Transactions**, **Payment Type**, and **Quality Score**, and prepared the project's central dataset for KPI calculation, customer analytics, and Tableau dashboard development.

Notebook: 05_merge_data – Data Integration and Final Dataset Preparation

This notebook combines all previously cleaned datasets into a single analytical dataset for KPI calculation, customer analysis, and Tableau dashboard development.

Main implementation steps:

- Imported the required libraries and loaded all cleaned datasets:
    - **Contacts**

    - **Calls**
    - **Deals**
    - **Marketing Spend**
- Verified the structure, dimensions, and data types of each dataset before merging.
- Merged the CRM datasets using the corresponding unique identifiers while preserving data consistency and referential integrity.
- Validated the merge results by checking:
    - Number of matched and unmatched records;
    - Duplicate identifiers;
    - Missing values introduced during the merge process.
- Combined call activity with customer and deal information to enable comprehensive CRM analysis.
- Integrated marketing spend data to support marketing performance evaluation and KPI calculations.
- Performed final data cleaning after the merge:
    - Removed unnecessary columns;
    - Corrected remaining missing values where appropriate;
    - Optimized data types for analysis.
- Validated the final analytical dataset using descriptive statistics and quality checks.
- Exported the completed analytical dataset in formats suitable for further analysis and Tableau dashboard development.

Result

A unified analytical dataset was created by successfully integrating all project data sources. The dataset provides a complete view of marketing activities, customer interactions, sales performance, and financial metrics, serving as the primary data source for KPI calculations and interactive dashboard development.

Notebook: 06_tableau_export – Tableau Data Preparation

This notebook prepares the final dataset for visualization in Tableau by generating additional calculated fields, optimizing the dataset structure, and exporting the required files.

Main implementation steps:

- Loaded the final analytical dataset.
- Verified the integrity of all calculated metrics before export.

- Created additional analytical fields required for Tableau dashboards, including business KPIs and supporting dimensions.
- Optimized data types and column structure to improve Tableau performance.
- Generated datasets specifically designed for different dashboard components.
- Exported the final datasets in **CSV** format for Tableau.
- Performed a final validation to ensure compatibility between Python calculations and Tableau metrics.

Result

A Tableau-ready dataset was created with all required business metrics, optimized data types, and a structure suitable for interactive dashboard development. The exported files ensure consistent KPI calculations and efficient visualization performance within Tableau.

Final Project Result

Throughout the project, a complete CRM analytics pipeline was developed, covering every stage of the analytical workflow—from raw data preprocessing and cleaning to business metric calculation and interactive dashboard creation.

Project achievements:

- Standardized and cleaned multiple CRM data sources.
- Performed comprehensive data validation and quality assurance.
- Developed reusable utility functions to automate repetitive analytical tasks.
- Implemented business logic for calculating key CRM and financial metrics.
- Integrated multiple datasets into a unified analytical database.
- Prepared optimized datasets for Tableau visualization.
- Created an interactive Tableau dashboard for business performance monitoring and decision-making.

Technologies used

- **Python**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Jupyter Notebook**
- **Tableau**
- **Microsoft Excel**

Outcome

The project demonstrates the complete end-to-end workflow of a Data Analyst, including data preprocessing, data validation, feature engineering, KPI calculation, business analysis, and dashboard development. The resulting analytical solution enables effective monitoring of sales performance, marketing efficiency, customer behavior, and overall business performance through interactive visualizations and data-driven insights.